Integration

Chapter Overview

Recover a function from its derivative

Integrate powers and sums. Use a point on a curve to determine the constant of integration.

Calculate areas

Find areas between curves, lines and the x-axis. Split the calculation when a curve crosses the axis.

Evaluate definite integrals

Use upper and lower limits to calculate a numerical result.

Connect differentiation and integration

Understand why integration recovers an area from its rate of change.

Integrating x^n terms

Solution 3

Integration reverses differentiation.

$5x^3 \xrightarrow{\text{differentiate}} 15x^2$

The power rule

$$\int ax^n \, dx = \frac{ax^{n+1}}{n+1} + c \qquad (n \neq -1)$$

Increase the power by one, then divide by the new power.

If $dy/dx = 3x^2$, find y.

Answer

$y = x^3 + c$

The derivative of every constant is zero. The constant c represents the whole family of possible functions.

Examples

Solution 4

Find y for each derivative. Include the constant of integration.

1.

$$\frac{dy}{dx} = 4x^3$$

1

$y = x^4 + c$

2.

$$\frac{dy}{dx} = x^5$$

2

$y = \frac{x^6}{6} + c$

3.

$$\frac{dy}{dx} = 3x^{1/2}$$

3

$y = 2x^{3/2} + c$

Check by differentiating your answer. Fractional and negative powers obey the same rule, except for power -1.

More Fractional Examples

Solution 5

Find y for each derivative. Include the constant of integration.

1. $\frac{dy}{dx} = \frac{4}{\sqrt{x}}$

1
$y = 8x^{1/2} + c$

2. $\frac{dy}{dx} = 5x^{-2}$

2
$y = -5x^{-1} + c$

3. $\frac{dy}{dx} = 4x^{2/3}$

3
$y = \frac{12}{5}x^{5/3} + c$

4. $\frac{dy}{dx} = 10x^{-2/7}$

4
$y = 14x^{5/7} + c$

Check by differentiating your answer. Fractional and negative powers obey the same rule, except for power -1.

Test Your Understanding

Solution 6

Find f(x) from each derivative. For 3 and 4, reveal the rewrite before the integral.

1. $f\prime(x) = 2x + 7$

1: integral
$f(x) = x^2 + 7x + c$

2. $f\prime(x) = x^2 - 1$

2: integral
$f(x) = \frac{x^3}{3} - x + c$

3. $f\prime(x) = \frac{2}{x^7}$

3: rewrite
$f\prime(x) = 2x^{-7}$

3: integral
$f(x) = -\frac{1}{3}x^{-6} + c$

4. $f\prime(x) = \sqrt[3]{x}$

4: rewrite
$f\prime(x) = x^{1/3}$

4: integral
$f(x) = \frac{3}{4}x^{4/3} + c$

5. $f\prime(x) = 33x^{5/6}$

5: integral
$f(x) = 18x^{11/6} + c$

The power rule excludes $x^{-1} = 1/x$: adding one would give a zero denominator. Its integral is studied in Year 2.

Exercise 13A – Y1 pages 289-290

Solution 7

4. Find f(x) when $f\prime(x) = (2x+3)^2$.

4: expand

$f\prime(x) = 4x^2 + 12x + 9$

4: integrate

$f(x) = \frac{4}{3}x^3 + 6x^2 + 9x + c$

5. Find f(x) when $f\prime(x) = 3x^{-2} + 6\sqrt{x} + x - 4$.

5: rewrite

$f\prime(x) = 3x^{-2} + 6x^{1/2} + x - 4$

5: integrate

$f(x) = -\frac{3}{x} + 4x^{3/2} + \frac{x^2}{2} - 4x + c$

Exercise 13A – Y1 pages 290

Solution 8

Challenge: integrate $\frac{(2\sqrt{x} - x^2)(3+x)}{x^5}$.

Expand and divide each term by x^5

$6x^{-9/2} + 2x^{-7/2} - 3x^{-3} - x^{-2}$

Integrate term by term

$-\frac{12}{7x^{7/2}} - \frac{4}{5x^{5/2}} + \frac{3}{2x^2} + \frac{1}{x} + c$

Integration notation

Solution 9

Notation for integration

$$\frac{d}{dx}(5x^2) = 10x \quad \Longleftrightarrow \quad \int 10x\,dx = 5x^2 + c$$

The integral sign

$\int$ tells us to integrate. An indefinite integral has no limits and represents a family of antiderivatives.

The variable

The dx tells us to integrate with respect to x. For example, $\int 6t^2\,dt = 2t^3 + c$.

Use brackets when the integrand contains several terms. Remember c for an indefinite integral.

Examples

Solution 10

Integrate each expression. In 3, p and q are constants.

1. $\int (x^{-3/2} + 2)\,dx$

1
$-2x^{-1/2} + 2x + c$

2. $\int (6t^2 - 1)\,dt$

2
$2t^3 - t + c$

3. $\int (px^3 + q)\,dx$

3
$\frac{px^4}{4} + qx + c$

Test Your Understanding — Solution 11

Given $y = 2x^5 + 6/\sqrt{x}$, where x > 0, find $\int y\,dx$.

Write the root as a power

$y = 2x^5 + 6x^{-1/2}$

Integrate

$\int y\,dx = \frac{x^6}{3} + 12\sqrt{x} + c$

Exercise 13B – Y1 pages 291-293 — Solution 12

Integrate each expression.

10. $\int \left(\frac{6}{x^2} + 4\sqrt{x} - 3x + 2\right) dx$

10

$-\frac{6}{x} + \frac{8}{3}x^{3/2} - \frac{3}{2}x^2 + 2x + c$

11. $\int \left(8x^3 + 6x - \frac{3}{\sqrt{x}}\right) dx$

11

$2x^4 + 3x^2 - 6\sqrt{x} + c$

Exercise 13B – Y1 pages 291-293 — Solution 13

12. (a) Write $(2+5\sqrt{x})^2$ as $4+k\sqrt{x}+25x$.

12a

$$(2+5\sqrt{x})^2 = 4+20\sqrt{x}+25x \Rightarrow k=20$$

(b) Hence integrate the expression.

12b

$$4x+\frac{40}{3}x^{3/2}+\frac{25}{2}x^2+c$$

13. Given $y=3x^5-4/\sqrt{x}$, find $\int y\,dx$.

13: rewrite

$$y=3x^5-4x^{-1/2}$$

13: integrate

$$\int y\,dx=\frac{x^6}{2}-8\sqrt{x}+c$$

Exercise 13B – Y1 pages 291-293 — Solution 14

14. Find p and q if $\int(\frac{p}{2x^2}+pq)\,dx=\frac{2}{x}+10x+c$.

14

$$-\frac{p}{2x}+pqx+c=\frac{2}{x}+10x+c$$

$$-\frac{p}{2}=2 \Rightarrow p=-4$$

$$pq=10 \Rightarrow q=-\frac{5}{2}$$

15. (a) Expand $(2-x)^{10}$, retaining terms up to x^2.

15a $(2-x)^{10}=1024-5120x+11520x^2+\cdots$

(b) Integrate these terms to obtain an approximation near x = 0.

15b

$$1024x-2560x^2+3840x^3+c$$

Finding constant of integration

Solution 15

Finding the constant of integration

All functions $y = x^3 + c$ have derivative $3x^2$. A known point selects one member of this family.

Find f(x) if $f\prime(x) = 3x^2$ and the curve passes through (1,3).

Integrate

$f(x) = x^3 + c$

Substitute the point

$3 = 1^3 + c \Rightarrow c = 2$

State the function

$f(x) = x^3 + 2$

Test Your Understanding

Solution 16

The curve has derivative $f\prime(x) = \frac{3}{8}x^2 - 10x^{-1/2} + 1$ for x > 0 and passes through (4,25).

(a) Find f(x).

a

$$f(x) = \frac{x^3}{8} - 20\sqrt{x} + x + c$$

$$25 = 8 - 40 + 4 + c \Rightarrow c = 53$$

$$f(x) = \frac{x^3}{8} - 20\sqrt{x} + x + 53$$

(b) Find the normal at (4,25).

b

$$f\prime(4) = 6 - 5 + 1 = 2$$

$$m_{\text{normal}} = -\frac{1}{2}$$

$$y - 25 = -\frac{1}{2}(x - 4)$$

$$x + 2y - 54 = 0$$

Exercise 13C – Y1 pages 294-295 — Solution 17

4. A curve passes through $(-1,0)$ and has $f\prime(x) = 9x^2 + 4x - 3$. Find f(x).

4

$$f(x) = 3x^3 + 2x^2 - 3x + c$$

$$0 = -3 + 2 + 3 + c \Rightarrow c = -2$$

$$f(x) = 3x^3 + 2x^2 - 3x - 2$$

5. Find y if $dy/dx = 3x^{-1/2} - 2x\sqrt{x}$ and y = 10 when x = 4.

5

$$y = 6\sqrt{x} - \frac{4}{5}x^{5/2} + c$$

$$10 = 12 - \frac{128}{5} + c \Rightarrow c = \frac{118}{5}$$

$$y = 6\sqrt{x} - \frac{4}{5}x^{5/2} + \frac{118}{5}$$

Exercise 13C – Y1 pages 294-295 — Solution 18

6. A curve has $dy/dx = (6x + 5x^{3/2})/\sqrt{x}$ and y = 100 at x = 9.

(a) Write the derivative as $6x^p + 5x^q$.

(b) Find the equation of the curve.

a

$$\frac{dy}{dx} = 6x^{1/2} + 5x$$

$$p = \frac{1}{2}, \quad q = 1$$

b

$$y = 4x^{3/2} + \frac{5}{2}x^2 + c$$

$$100 = 108 + \frac{405}{2} + c$$

$$y = 4x^{3/2} + \frac{5}{2}x^2 - \frac{421}{2}$$

Definite Integration

Solution 19

Definite integration

A definite integral has lower and upper limits. Evaluate an antiderivative at the upper limit, then subtract its value at the lower limit.

$$\int_1^5 4x^3\,dx$$

Example

$$\left[x^4\right]_1^5 = 5^4 - 1^4 = 624$$

Connection with area

For a nonnegative function, the definite integral gives the area under the graph between the limits. For a speed–time graph, this area represents distance travelled.

Another Example

Solution 20

Evaluate $\int_{-3}^{3}(x^2+1)\,dx$.

Integrate and substitute both limits

Answer

$$\begin{aligned}\int_{-3}^{3}(x^2+1)\,dx &= \left[\frac{x^3}{3}+x\right]_{-3}^{3}\\ &= (9+3)-(-9-3)\\ &= 24\end{aligned}$$

Why is there no c?

$$[F(x)+c]_a^b = F(b)+c-F(a)-c$$

The constants cancel. Use brackets around the whole lower-limit value, especially when it is negative.

Problem Solving

Solution 21

Find P if $\int_1^5 (2Px + 7)\,dx = 4P^2$.

$[Px^2 + 7x]_1^5 = 24P + 28$

$4P^2 = 24P + 28$

$P^2 - 6P - 7 = (P - 7)(P + 1) = 0$

$P = 7 \quad \text{or} \quad P = -1$

Exercise 13D – Y1 page 297

Solution 22

4. Find A if $\int_1^4 (6\sqrt{x} - A)\,dx = A^2$.

4

$[4x^{3/2} - Ax]_1^4 = 28 - 3A$

4

$A^2 + 3A - 28 = (A + 7)(A - 4) = 0$

$A = -7 \quad \text{or} \quad A = 4$

Exercise 13D – Y1 page 297 — Solution 23

5. Evaluate $\int_1^9 (2x - 3\sqrt{x})\,dx$.

6. Express $\int_4^{12} \frac{2}{\sqrt{x}}\,dx$ in the form a + b$\sqrt{3}$.

5

$$[x^2 - 2x^{3/2}]_1^9$$
$$= (81 - 54) - (1 - 2) = 28$$

6

$$[4\sqrt{x}]_4^{12} = 4\sqrt{12} - 8$$
$$= -8 + 8\sqrt{3}$$

Exercise 13D – Y1 page 297 — Solution 24

7. Find k if $\int_1^k \frac{1}{\sqrt{x}}\,dx = 3$.

7

$$[2\sqrt{x}]_1^k = 2\sqrt{k} - 2 = 3$$

7

$$\sqrt{k} = \frac{5}{2} \Rightarrow k = \frac{25}{4}$$

EXTENSION

Solution 25

1. [MAT 2009 1A] Find the smallest value of $I(a) = \int_0^1 (x^2 - a)^2\,dx$ as a varies.

1

$$I(a) = \frac{1}{5} - \frac{2}{3}a + a^2$$

$$= \left(a - \frac{1}{3}\right)^2 + \frac{4}{45}$$

The minimum is $4/45$, attained when $a = 1/3$.

2. [MAT 2015 1D] Let $f(x) = \int_0^1 (xt)^2\,dt$ and $g(x) = \int_0^x t^2\,dt$. For A > 0, which is greater: g(f(A)) or f(g(A))?

2

$$f(x) = \frac{x^2}{3}, \qquad g(x) = \frac{x^3}{3}$$

$$g(f(A)) = \frac{A^6}{81}, \quad f(g(A)) = \frac{A^6}{27}$$

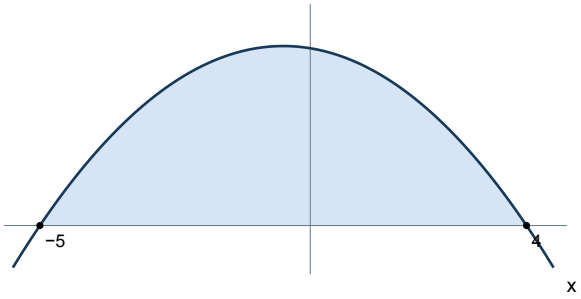
Therefore $f(g(A)) > g(f(A))$ for every A > 0.

Areas under curves

Solution 26

For a curve above the x-axis, its area between x = a and x = b is

$$\int_a^b f(x)\,dx$$



Find the finite area bounded by $y = 20 - x - x^2$ and the x-axis.

Limits

$$20 - x - x^2 = (4 - x)(x + 5) = 0$$

The limits are -5 and 4.

Area

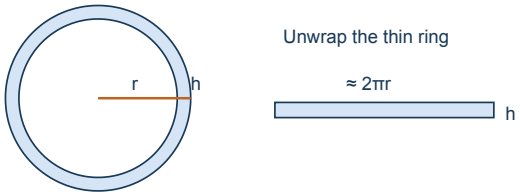
$$\int_{-5}^{4} (20 - x - x^2)\,dx$$

$$= \left[20x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-5}^{4} = \frac{243}{2}$$

Just for your interest... Solution 27

Why does integration give area?

A circle has area $A = \pi r^2$, so $dA/dr = 2\pi r$, its circumference.



Increase the radius by a small amount h.

$\pi(r+h)^2 - \pi r^2 = 2\pi rh + \pi h^2$

The thin ring is approximately a strip of length 2πr and width h.

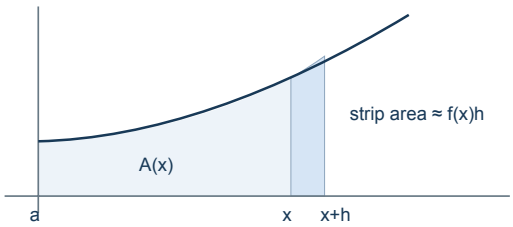
Divide by h and let h approach zero.

$\dfrac{\Delta A}{h} = 2\pi r + \pi h \longrightarrow 2\pi r$

Adding these thin rings is integration: it recovers the area from its rate of change.

Why integration gives area Solution 28

For a sphere, $V = \frac{4}{3}\pi r^3$, so $dV/dr = 4\pi r^2$, its surface area.



For a graph, let A(x) be the accumulated area from a fixed starting point to x.

A thin strip of width h has approximate area $f(x)h$.

$\dfrac{A(x+h) - A(x)}{h} \longrightarrow f(x)$

Therefore differentiation of accumulated area gives the original function.

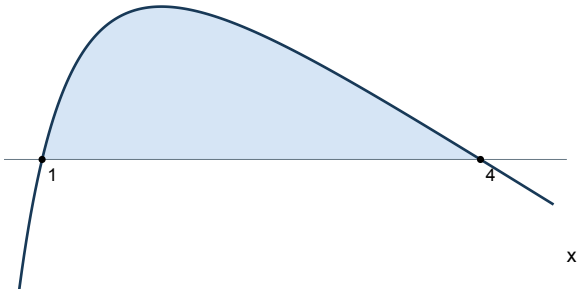
$A\prime(x) = f(x)$

$\displaystyle\int_a^b f(x)\,dx = A(b) - A(a)$

Test Your Understanding

Solution 29

Find the area under $y = 27 - 2x - 9\sqrt{x} - 16/x^2$ between x = 1 and x = 4.



Set up

$$\int_1^4 (27 - 2x - 9x^{1/2} - 16x^{-2})\,dx$$

Evaluate

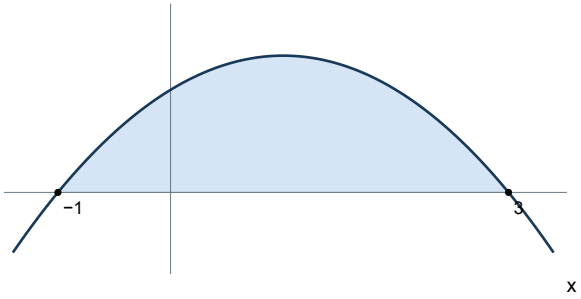
$$= \left[27x - x^2 - 6x^{3/2} + \frac{16}{x}\right]_1^4$$

$$= 48 - 36 = 12$$

Exercise 13E – Y1 pages 299-300

Solution 30

8. The curve $y = -x^2 + 2x + 3$ bounds a finite region with the x-axis. Find its area.



Limits

$$-x^2 + 2x + 3 = (3 - x)(x + 1) = 0$$

The intercepts are $(-1, 0)$ and $(3, 0)$.

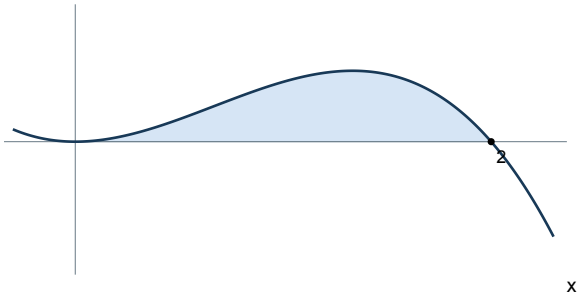
Area

$$\int_{-1}^{3} (-x^2 + 2x + 3)\,dx$$

$$= \left[-\frac{x^3}{3} + x^2 + 3x\right]_{-1}^{3} = \frac{32}{3}$$

Exercise 13E – Y1 pages 299-300 Solution 31

9. Find the finite area bounded by $y = x^2(2 - x)$ and the x-axis.



Limits

The roots are $x = 0$ and $x = 2$; the curve is nonnegative between them.

Area

$$\int_0^2 (2x^2 - x^3)\,dx$$

$$= \left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = \frac{4}{3}$$

EXTENSION Solution 32

1. [MAT 2007] Given

$$3\int_0^1 f(x)\,dx + 2\int_1^2 f(x)\,dx = 7$$

$$\int_0^2 f(x)\,dx + \int_1^2 f(x)\,dx = 1$$

find $\int_0^2 f(x)\,dx$.

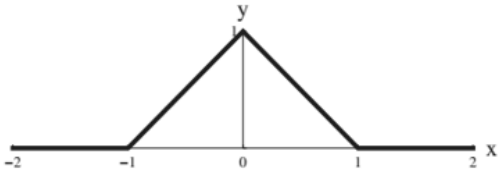
1

Let $a = \int_0^1 f$ and $b = \int_1^2 f$.

$3a + 2b = 7, \quad a + 2b = 1$

$a = 3, \quad b = -1, \quad a + b = 2$

2. [MAT 2011] Using the graph of f, evaluate $\int_{-1}^1 f(x^2 - 1)\,dx$.



2

For $-1 \le x \le 1$, we have $-1 \le x^2 - 1 \le 0$. On this interval $f(t) = t + 1$.

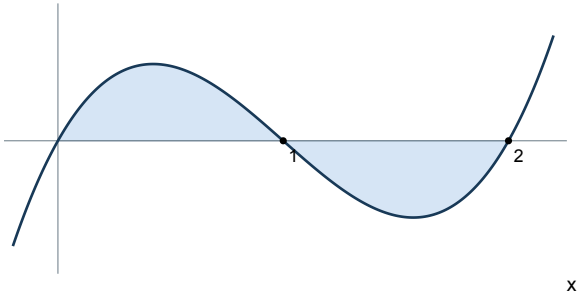
$f(x^2 - 1) = x^2$

$$\int_{-1}^1 x^2\,dx = \frac{2}{3}$$

‘Negative Areas’ Solution 33

Definite integrals give signed area.

Area above the x-axis contributes positively. Area below it contributes negatively.



Consider $f(x) = x(x-1)(x-2)$.

Does zero mean there is no area?

Integral

$$\int_0^2 (x^3 - 3x^2 + 2x)\,dx$$

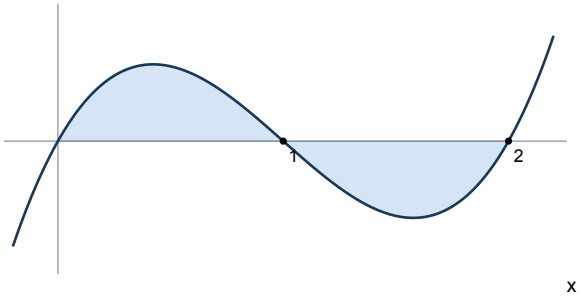
$$= \left[\frac{x^4}{4} - x^3 + x^2\right]_0^2 = 0$$

Meaning

No. The area above the axis on $0 \leq x \leq 1$ cancels the signed area below it on $1 \leq x \leq 2$. Geometric area is always nonnegative.

Example Solution 34

Find the total area between $y = x(x-1)(x-2)$ and the x-axis from $x = 0$ to $x = 2$.



Above the axis

$$\int_0^1 f(x)\,dx = \frac{1}{4}$$

Below the axis

$$\int_1^2 f(x)\,dx = -\frac{1}{4}$$

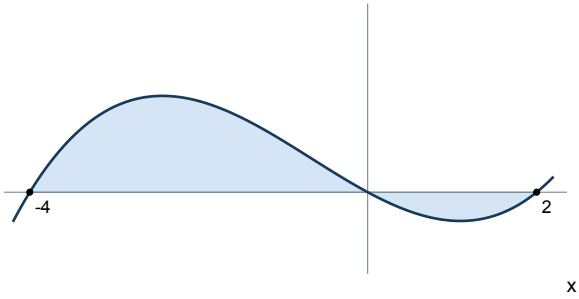
Total

$$\text{Total area} = \frac{1}{4} + \left|-\frac{1}{4}\right| = \frac{1}{2}$$

Test Your Understanding

Solution 35

The curve is $y = x(x+4)(x-2)$. (a) Find its x-intercepts. (b) Find the total area of the finite regions bounded by the curve and the x-axis.



a $x = -4,\ 0,\ 2$

Integrate

$$F(x) = \frac{x^4}{4} + \frac{2x^3}{3} - 4x^2$$

Split at zero

$$\int_{-4}^{0} y\,dx = \frac{128}{3}$$

$$\int_{0}^{2} y\,dx = -\frac{20}{3}$$

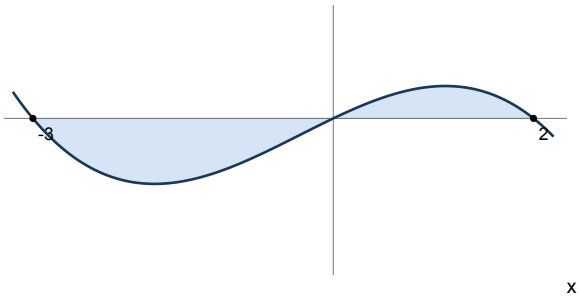
b

$$\text{Total area} = \frac{148}{3}$$

Exercise 13F – Y1 pages 301-302

Solution 36

2. The curve is $y = x(x+3)(2-x)$. (a) Find its x-intercepts. (b) Find the total area of the finite regions bounded by the curve and the x-axis.



a $x = -3,\ 0,\ 2$

Integrate

$$F(x) = -\frac{x^4}{4} - \frac{x^3}{3} + 3x^2$$

Split at zero

$$\int_{-3}^{0} y\,dx = -\frac{63}{4}$$

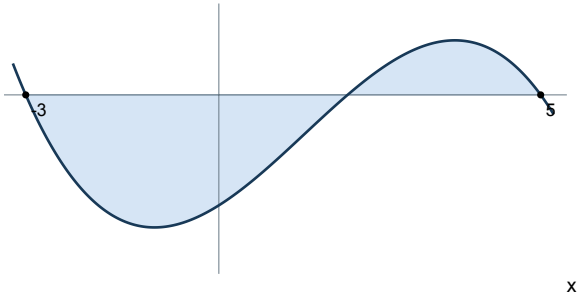
$$\int_{0}^{2} y\,dx = \frac{16}{3}$$

b

$$\text{Total area} = \frac{253}{12}$$

Exercise 13F – Y1 pages 301-302 Solution 37

3. Let $f(x) = -x^3 + 4x^2 + 11x - 30$. Show that x + 3 is a factor, factorise fully, find all roots, and calculate the total area between the curve and the x-axis.



Factorise

$f(-3) = 27 + 36 - 33 - 30 = 0$

$f(x) = (x+3)(-x^2 + 7x - 10)$

$= (x+3)(2-x)(x-5)$

$x = -3,\ 2,\ 5$

Area

$$F(x) = -\frac{x^4}{4} + \frac{4x^3}{3} + \frac{11x^2}{2} - 30x$$

$$\int_{-3}^{2} f(x)\,dx = -\frac{1375}{12}$$

$$\int_{2}^{5} f(x)\,dx = \frac{117}{4}$$

$$\text{Total} = \frac{1375}{12} + \frac{117}{4} = \frac{863}{6}$$

EXTENSION Solution 38

[MAT 2010] For a > 0, where is $I(a) = \int_0^a (4 - 2^{x^2})\,dx$ stationary?

Answer

$I\prime(a) = 4 - 2^{a^2} = 0$

$2^{a^2} = 2^2 \Rightarrow a = \sqrt{2}$

[STEP 2014 Q3] Let b > a ≥ 0 and

$$\int_a^b x^2\,dx = \left(\int_a^b x\,dx\right)^2$$

(i) Find b when a = 0. (ii) When a = 1, show that b satisfies a cubic with exactly one real root, between 2 and 3. (iii) Put p = b + a and q = b − a. Show that $3p^2 + q^2 = 3p^2q$ and deduce $1 < q \leq 4/3$.

Worked solution on the next slide.

Guidance for Extension Problem 2

Solution 39

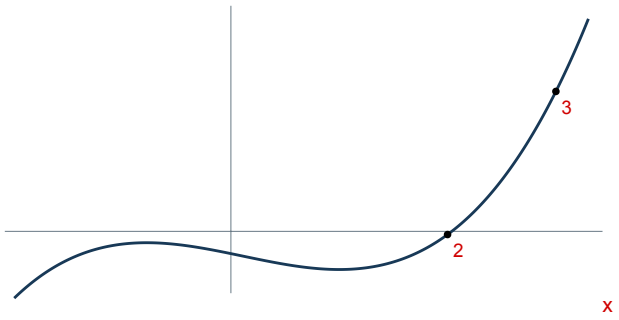
(i) a = 0

(ii) a = 1

i

$$\frac{b^3}{3} = \frac{b^4}{4} \Rightarrow b = \frac{4}{3}$$

ii: sketch



ii

$$\frac{b^3 - 1}{3} = \frac{(b^2 - 1)^2}{4}$$

Divide by b − 1, since b > 1:

$$3b^3 - b^2 - 7b - 7 = 0$$

For h(b) equal to this cubic, h(2) = −1 and h(3) = 44. Its turning points at −7/9 and 1 both have negative values, so it has exactly one real root, between 2 and 3.

(iii) p = b + a, q = b − a

iii

$$\frac{q(3p^2 + q^2)}{12} = \frac{p^2 q^2}{4}$$

$$3p^2 + q^2 = 3p^2 q, \qquad p^2 = \frac{q^2}{3(q-1)}$$

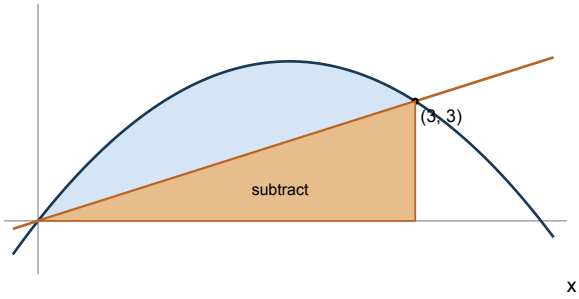
Positivity gives q > 1. Since a ≥ 0, p ≥ q, so

$$\frac{1}{3(q-1)} \geq 1 \Rightarrow 1 < q \leq \frac{4}{3}$$

Areas between curves and lines

Solution 40

Find the area between $y = x(4 - x)$ and $y = x$.



1. Intersections

$$x(4 - x) = x \Rightarrow x(3 - x) = 0$$

The intersections have x = 0 and x = 3.

2. Area under curve

$$\int_0^3 (4x - x^2)\, dx = 9$$

3. Subtract triangle

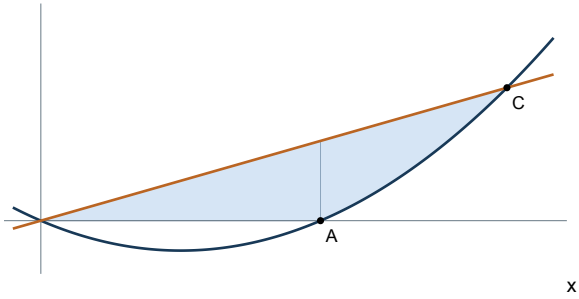
$$\text{Triangle} = \frac{1}{2} \times 3 \times 3 = \frac{9}{2}$$

$$\text{Required area} = 9 - \frac{9}{2} = \frac{9}{2}$$

A Harder One

Solution 41

The curve is $y = x(x-3)$ and the line is $y = 2x$. Find the region above the x-axis, below the line and to the left of the positive branch of the curve.



1. Points

$x(x-3) = 2x \Rightarrow x = 0, 5$

C = (5, 10); the positive x-intercept of the curve is A = (3, 0).

2. Strategy

Use the triangle with vertices (0, 0), (5, 0), (5, 10), then subtract the area under the curve from 3 to 5.

3. Area

$\text{Triangle} = \frac{1}{2}(5)(10) = 25$

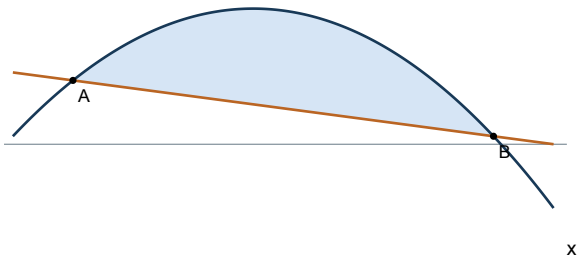
$$\int_3^5 (x^2 - 3x)\,dx = \frac{26}{3}$$

$$\text{Required area} = 25 - \frac{26}{3} = \frac{49}{3}$$

Test Your Understanding

Solution 42

The curve $y = 10x - x^2 - 8$ and line $y = 10 - x$ enclose a region. (a) Find their intersections. (b) Find the area.



a

$10x - x^2 - 8 = 10 - x$

$x^2 - 11x + 18 = (x-2)(x-9) = 0$

The points are A = (2, 8), B = (9, 1).

b

$$\int_2^9 \left[(10x - x^2 - 8) - (10 - x)\right] dx$$

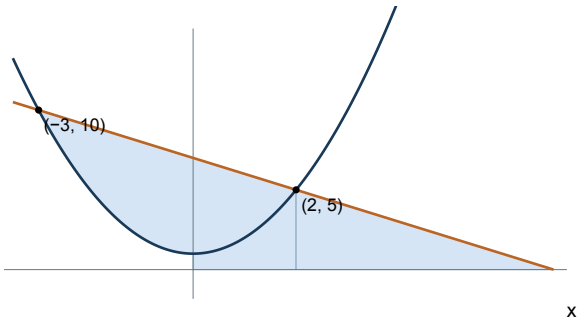
$$= \left[\frac{11x^2}{2} - \frac{x^3}{3} - 18x\right]_2^9 = \frac{343}{6}$$

General rule:

$$\text{Area} = \int_a^b (\text{upper function} - \text{lower function})\,dx$$

Exercise 13G – Y1 pages 304-306 Solution 43

6. The curve $y = x^2 + 1$ and line $y = 7 - x$ enclose R_1. R_2 is the region in the first quadrant below both graphs. Find both areas.



R_1

$x^2 + 1 = 7 - x \Rightarrow (x+3)(x-2) = 0$

$$R_1 = \int_{-3}^{2} (6 - x - x^2)\,dx = \frac{125}{6}$$

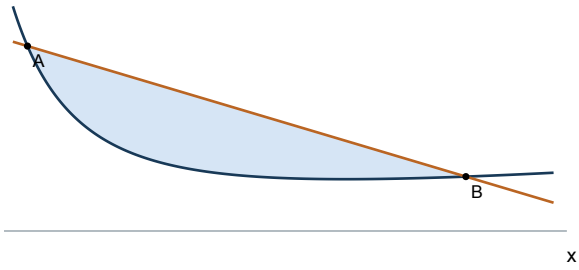
R_2

$$R_2 = \int_0^2 (x^2+1)\,dx + \tfrac{1}{2}(5)(5)$$

$$= \frac{14}{3} + \frac{25}{2} = \frac{103}{6}$$

Exercise 13G – Y1 pages 304-306 Solution 44

8. Points A and B on $y = 2/x^2 + x$ have x-coordinates $1/2$ and 2. Find the area between chord AB and the curve, to 3 significant figures.



Line

$A = (\tfrac{1}{2}, \tfrac{17}{2}), \quad B = (2, \tfrac{5}{2})$

$$m = \frac{5/2 - 17/2}{2 - 1/2} = -4$$

$$AB:\ y = \frac{21}{2} - 4x$$

Area

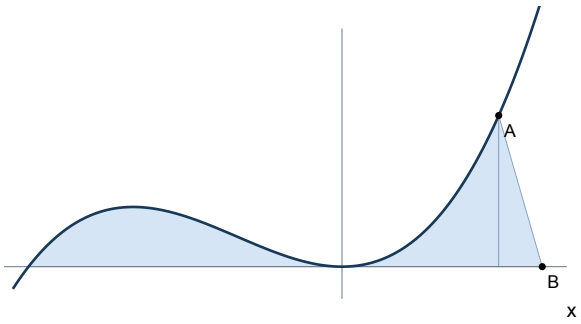
$$\int_{1/2}^{2} \left(\frac{21}{2} - 5x - \frac{2}{x^2}\right) dx$$

$$= \left[\frac{21x}{2} - \frac{5x^2}{2} + \frac{2}{x}\right]_{1/2}^{2}$$

$$= \frac{27}{8} \approx 3.38$$

Exercise 13G – Y1 pages 304-306 — Solution 45

10. The curve is $y = x^2(x+4)$. R_1 lies under the curve between -4 and 0. R_2 lies under the curve from 0 to A = (2, 24), then under the straight segment from A to B = (b, 0). Find b if the areas are equal.



R_1

$$R_1 = \int_{-4}^{0} (x^3 + 4x^2)\,dx = \frac{64}{3}$$

R_2 and b

$$R_2 = \int_{0}^{2} (x^3 + 4x^2)\,dx + \tfrac{1}{2}(b-2)(24)$$

$$= \frac{44}{3} + 12(b-2)$$

$$\frac{64}{3} = -\frac{28}{3} + 12b \Rightarrow b = \frac{23}{9}$$

EXTENSION — Solution 46

a. [MAT 2005] Find the area between $y = x^2$ and $y = x + 2$.

a

$$x^2 = x + 2 \Rightarrow x = -1, 2$$

$$\int_{-1}^{2} (x + 2 - x^2)\,dx = \frac{9}{2}$$

b. [MAT 2016] Let $f(x) = a - x^2$ and $g(x) = x^4 - a$, with a > 0. When is the finite area bounded by f and the x-axis greater than the corresponding area for g?

b

$$A_1 = \int_{-\sqrt{a}}^{\sqrt{a}} (a - x^2)\,dx = \frac{4}{3}a^{3/2}$$

$$A_2 = \left| \int_{-a^{1/4}}^{a^{1/4}} (x^4 - a)\,dx \right| = \frac{8}{5}a^{5/4}$$

$$A_1 > A_2 \iff a^{1/4} > \frac{6}{5} \iff a > \left(\frac{6}{5}\right)^4$$